

Sprint 1 (Feb 12 - Feb 23):

Amelia:

- Studied background readings to learn about the context of the project
- Reviewed the Project Description document
- Investigated the datasets and data dictionary
- Took a closer look at “[Canopy Change Assessment: Tree Canopy Change Metrics](#)” in ArcGIS online viewer to better understand the different layers and to think about how these can be used to answer the questions outlined in the project description
 - There was no CSV available so there were limited capabilities for analysis at this point.

Grace:

- Read background readings, project description, look through data sets and data dictionaries, and see how the questions we are going to answer will be relevant.
- Previewed [Canopy Change Assessment: Tree Canopy Metrics](#) and the geojson files from arcGIS Hubset. Also explored on google colab.
- [Notion Card](#)
- Helped create presentation formatting, and added questions to the slides.
- Looked into ArcGIS.

Gui:

- Read and caught up on all project content including project overview, who the client is, the scope of this project, the broader/long term implications and grasped an initial understanding of our task
- Analyzed and accessed all the datasets; tried some preliminary analysis with ArcGIS to visualize the datasets
 - Did not make any legitimate progress/analysis due to it being my first time on the software
- Given the number of initial datasets on the project description (6), we decided to assign each team member a dataset to analyze and learn. I was assigned the Canopy Change Assessment: Land Cover dataset. Along with the team, I then came up with a few questions to ask the clients for our first official client meeting.
- My only blocker at this time was a logistical issue relating to connecting my BU email to ArcGIS and being able to have access to ArcGIS’ full capabilities so I could upload and analyze the datasets. I fixed this relatively quickly
- [Notion Cards](#)

Rachel:

- Read through the project description and background documents to understand scope of the questions and the importance of tree canopy to the community
- Explored the potential uses of the [Tree Canopy Metrics Dataset](#) by going through all layers of the file, and understanding how they aggregated tree canopy differently using different aggregations of land area.
 - Also reviewed column names to understand which would be useful to calculating tree canopy metrics.

- Link to Notion Card of write up: [HERE](#)
- Also helped to create questions for the client alongside the rest of the team for the kickoff meeting

Talya:

Sprint 2 (Feb 24 - March 10):

Amelia:

- Out due to medical emergency - communicated this to team and continued to participate in team's weekly meetings
- Reviewed "[Tree Canopy Change Metrics: Calculate the Percentage of Tree Canopy by Land Use](#)" with Rachel

Grace:

- Initially I looked at "[Tree Canopy Change Metrics: Calculate the Percentage of Tree Canopy by Land Use](#)" for "What percentage of Boston's tree canopy is located on public land versus residential parcels?" with Rachel and Amelia.
- Was assigned to determine private tree canopy while Rachel did public tree canopy.
- Mapped Tree Canopy Polygons and Land Use into ArcGIS. Saw the same blocker as Rachel, where tree polygons occupied more than one land use.
- Then looked into [Tree Ordinance](#) and EJ Communities Data for : Are there inequities in tree canopy coverage between neighborhoods, particularly in marginalized communities?
- Also met with Amelia to understand her visualization on Looker Studio.

Gui:

- For our first client meeting, I used the Tree Canopy Metrics dataset to determine the tree canopy percentage in the city of Boston by neighborhood by doing the following:
 - 1) Downloaded and uploaded dataset on ArcGIS
 - 2) Activated the "NBHD % Tree Canopy Coverage" layer and opened the attribute table with all the relevant data in spreadsheet form
 - 3) Filtered the table to only include relevant columns such as TC_E_A (total tree canopy area), TC_Land_A (total land area), and neighborhood name
 - 4) Extracted to Excel and used a simple formula to find tree canopy percentage by neighborhood
- This process was relatively straightforward and the only blocker I faced was extracting the attribute table from ArcGIS into Excel to do the calculations. I was unable to do so and had to ultimately copy and paste the values manually.
- These results were presented on the first client meeting (03/05) which I helped present along with Talya (1st meeting at this point)
- [Notion Cards](#)

Rachel:

- For the first client meeting, I was assigned to answer: “What percentage of Boston's tree canopy is located on public land versus residential parcels?”, specifically to determine percentage on public parcels, while Grace was assigned to determine private tree canopy.
- At first I planned to use the [Tree Canopy Polygons](#) dataset and the [Land Use](#) dataset. At this time we were thinking of using an intersection between the Tree Canopy Polygons and the Land Use in ArcGIS because the Land Use dataset seemed to have a category named ‘residential’.
 - BLOCKER: At the time, we were blocked by not being able to classify tree canopy that occupied more than one land use since we were using a spatial join instead of an overlay intersect.
 - In addition the capabilities on ArcGIS Online were much less efficient than using ArcGIS pro, which at the time was our only option because ArcGIS Pro was not compatible with Macbooks.
- I tried using ‘join features’ in ArcGIS to overlay the land use polygons with the tree polygons, but I ended up with more records after the join. I think it duplicated any rows that intersected with more than one land use, but when I clicked on the polygons in map viewer it seemed to arbitrarily classify the polygons as one land use. Typically, it classified the entire tree area polygon as ROW, even if only a small fraction of the tree canopy intersected with it.
- For the client meeting, I prepared slides explaining my methodology at the time to calculate Public Tree Canopy as all land use classes, excluding residential, which would include Public Open Space, Institutional, ROW(Right-Of-Way), Industrial, Commercial, Mixed Use, Commercial2
 - Also created slide to check the way the client wanted to consider public vs residential as:
 - Tree locations - Public Open Space = Residential Trees OR
 - Tree locations - Residential = Public Trees
- After this client meeting it was clarified that we should not be using land use(“LU_GENERAL”) to determine public vs residential, instead it was clarified that public meant it was owned by the City of Boston and residential was considered to be everything not owned by the City of Boston.
 - So, we unblocked our issue by working with the Tree Canopy Metrics dataset instead which had tree canopy area already aggregated at a parcel level with a column that detailed the parcel owner.

Talya

Land Use Calculations:

- I calculated tree-canopy coverage for each land-use class using the Tree Canopy Change Metrics 2014–2019 layer from ArcGIS Online.
- I downloaded the attribute table from the layer and imported it into Excel for easier tabulation.
- I focused on two key columns:
 - **TC_E_A** – existing tree-canopy area (in square feet)
 - **TC_Land_A** – total land area, excluding water
- I grouped the data by the generalized land-use code (**LU_GENERAL**), such as Residential, Commercial, Institutional, Industrial, and Agricultural.
- For each land-use group, I: Summed **TC_E_A** and **TC_Land_A**
- Calculated canopy coverage using the formula: **Canopy % = (sum of TC_E_A ÷ sum of TC_Land_A) × 100**

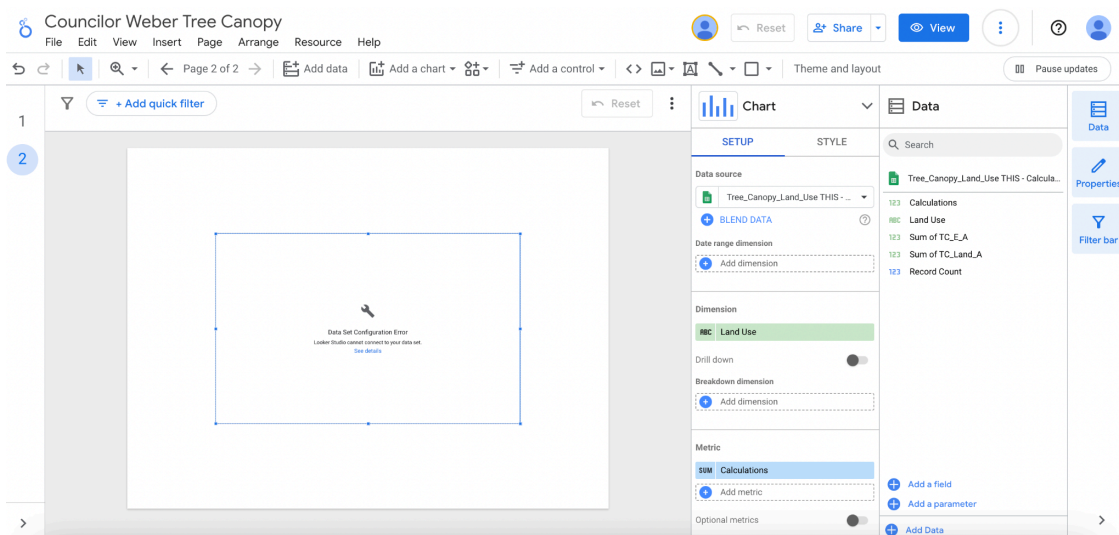
- I used summed areas to ensure large parcels (like industrial lots) had a proportionate impact on the result.
- I created a PivotTable with a calculated field to keep the process transparent and easy to update when new data is available.
- The final percentages show how much canopy exists in each land-use category, helping identify where coverage is high, low, or could be improved.

Visualization 1:

- I uploaded the saved *Tree_Canopy_Land_Use.xlsx* to Google Drive and opened it with Google Sheets so Looker Studio could read it.
- I made a blank sheet in Looker Studio, added the data, and selected the workbook tab within the data.
- I verified the field types to ensure this would not cause issues within the visualization.
- Added a bar chart to the sheet.
- Added Neighborhood (Name) to Dimension.
- Added the existing Tree Canopy Percentage into the Metric as Average using the calculations I previously did.
- I sorted the bars in descending order to put land-use classes with the highest coverage first.
- Within the style tab, I adjusted the colors, added a title, and made sure all necessary numbers were shown within the chart.

Amelia and Talya:

- Blocker: Could not connect to the data source when trying to make a visualization
- Tried to refresh the data source and Looker Studio



Sprint 3 (:

Rachel

- Worked on our in class presentation during class to prepare a three slide presentation summarizing the background purpose of our project, our strategy, and updates on current progress.
- After the client meeting, I worked on:

- Used Owner in the Tree Canopy Metrics dataset instead to determine Public vs Residential based on the owner column.
- Uploaded a preliminary [Jupyter Notebook](#) to the Drive exploring the dataset
- BLOCKER: Found that there are over 100k unique owners. CITY OF BOSTON, CITY OF BOSTON BY FCL, and COMMONWEALTH OF MASS were the most common, but had to consider what other owner names would be considered public
- Reviewed tree ordinance document sent by Jordan to determine names of parcel owners that are considered owned by city of Boston:
 - City of Boston
 - Boston Water and Sewer Commission
 - Boston Housing Authority
 - Boston Public Schools*
 - Boston Redevelopment Authority
 - Boston Planning and Development Agency
- BLOCKER: Need to determine a way to match for typos in the owners
- Pushed code to Github answering q1 “What percentage of Boston's tree canopy is located on public land versus residential parcels?” using the methods above.
 - Used string searches to create a list of owners considered protected by the tree ordinance: kept using outputs of string searches and saving them to a list
 - Used Fuzzy Matching to determine close matches to the owners listed above in the ordinance
 - Used Property Occupancy Codes (“PTYPE”): used codes that started with a 9 to filter through tax exempt property that were most likely to be owned by the state or city
 - Met with Tom Hopper, an expert in residence, to understand what PTYPE’s represent and how they could be utilized in this analysis for determining all publicly owned properties, such as state and federally owned properties
 - Calculated the percentage as **Percentage of Tree Canopy that is Public or Private/Total Existing Tree Canopy in Boston** to understand the distribution of tree canopy in Boston.
- Also pushed code answering the following questions to Github:
 - “What are the trends in tree canopy loss or gain (2014-2019) for public and private areas?”
 - Used the Tree Canopy Change dataset and grouped all parcels by their ownership as public, private, or ROW, by summing over the columns “Gain” and “Loss” to calculate the gain and loss for all public and private parcels.
 - “Which areas have the highest potential for expanding tree canopy, and are they public or private spaces?”
 - Used the Tree Canopy Metrics dataset’s column “TC_P_A”, which represents possible area for tree canopy to aggregate the potential canopy for public and private parcels.
 - Grouped by ownership of the parcel to sum the potential tree canopy
- Reviewed EJ Communities Dataset: Are there inequities in tree canopy coverage between neighborhoods, particularly in marginalized communities?

- Found that there are 33 discrepancies in GIS ID's for census tracts between the tree canopy metrics dataset and EJ communities datasets
- Worked to create slides 6 -14 for the mid semester client deck answering the questions described above.
- Worked on creating visualization in looker studio
 - [Visualization 3 in Looker Studio for Tree Canopy Coverage by Public vs Private](#)
 - [Visualization 4 in Looker Studio for Tree Canopy Gain and Loss](#)
 - Got familiar with how to use looker studio and worked to use python to make the data easier to use for visualizing

Amelia

- Worked on "[Visualization 1 in Looker Studio for Tree Canopy Coverage by Neighborhood](#)" with Talya in Looker Studio
 - Explored Looker Studio as this was my first time using it
 - Created visualization with client's perspective in mind; played around with different formatting for the bar graph such as colors, fonts, and a layout that would most effectively convey the information as clearly as possible to the clients to help them make informed decisions
- Supported Talya in the creation of Visualization 2; Ran into blocker trying to connect data source

Grace:

- Because my portion of the task of determining private tree canopies was easily answered with Rachel's task of public tree canopies, I decided to tackle the EJ community question.
- Mapped EJ community dataset and CT dataset into ArcGIS.
- Blocker: saw that there were missing slivers of land in CT that EJ dataset had.
- Unblocked by discovering that the missing land was most likely due to not being residential/no census tract.

Gui:

- Fixed Amelia and Talya's blocker on connecting Excel datasets to Looker Studio to create visualizations
- Worked on "[Visualization 2 in Looker Studio for Tree Canopy Coverage by Land Use](#)"
 - Faced some initial blockers on how to use Looker Studio given it was my first time using this platform, but overcame these by watching youtube videos and asking ChatGPT for instructions
 - Visualization included a bar chart with % tree canopy on the y-axis and type of land use on the x-axis
- Hosted another client presentation (2nd at this point) along with Talya to present team's progress thus far

Sprint 4 (- April 20):

Amelia

- Submitted resource request for team to get access to BigQuery
 - Gained access to BigQuery for this project

- Presented Question 6: “Are there inequities in tree canopy coverage between neighbors particularly in marginalized communities?” to client during mid-semester client meeting
 - Outlined the current progress made to answer this question including trying to merge boston_gdf, CT_tree_metrics, and Ej_tract
 - Presented blocker of Ej_tract missing 75/255 tracts
- Gave an overview of blockers being faced at this point
 - May be an issue with how many data sources/size of the sheet you can connect to Looker Studio because it's free.
 - File itself is unable to be read by Looker Studio
 - We were able to move through this blocker and create the second visualization.

Grace:

- Instead of using ArcGIS directly due to limited experience, resorted to geopandas in order to explore and look into the dataset.
- Had a call and working session with Rachel in order to see how I could approach this using geopandas.
- Continued to work on my portion as well as understand other's work
- Work week (more exploration and testing)
- Organized mid semester client meeting with group

Rachel:

- Explored using the [Citywide Land Audit](#) to distinguish between publicly owned and privately owned trees: [Notion](#)
 - I used python to explore how the parcels of the Citywide Land Audit were different from the tree canopy metrics dataset
 - The tree canopy metrics dataset was created in 2019 using 2019 parcels, but the Citywide Land Audit used parcels from 2022 which were different.
 - There are discrepancies most likely because of the property ownership differences between 2019 and 2022. For example, below the parcel was owned by Massachusetts port in 2019, but in the 2022 land audit it appears to be under the care of the Boston Planning and Development Agency.
 - In addition, there are 360 parcels found by Fuzzy Matching that are not in the land audit, even though they have owners like “City of Boston”. This again might be due to parcels changing owners.
 - Only 34 parcels in the 2022 land audit don't exist in the 2019 tree canopy metrics. So the discrepancies above are most likely due to ownership changing rather than mismatches in parcelization.
 - We dropped one of these parcels because it was only 9-digits long, which seems to be a data entry error because they're supposed to be 10-digits long
 - This code exploring the data is in the Jupyter Notebook named Analysis.ipynb in Github
 - In ArcGIS Pro, I used the tree canopy polygons dataset and the Citywide Land Audit dataset to recalculate the area of tree canopy that overlapped with the city audit parcels, which I considered city-owned. Then I used a Polygon from the 2019 tree canopy metrics dataset to use the overlay intersect tool to calculate all tree canopy in the right-of-way. I considered any remaining tree canopy non-city owned.

- Despite being able to recalculate the percentage of total tree canopy, it was not possible to recalculate the gain/loss or potential tree canopy area in this way because the geospatial data is not available on Analyze Boston
- I emailed Analyze Boston to ask if the Gain/Loss dataset could be made available
 - The file turned out to be a .tif file that needed to be converted to a geojson or shapefile in ArcGIS. However, at the time I was unfamiliar with how the file type could be used.
- We chose to continue using FuzzyMatching to aggregate metrics instead due to these limitations.
- Notion Percentages Calculated: [Notion](#)

Talya:

Visualization 2 fix:

- I noticed only 10 neighborhoods were shown on the bar chart, and I wanted to represent all 17 neighborhoods.
- Under style, I adjusted the number of bars to be shown and adjusted the chart so that all neighborhood names are readable.

Dashboard:

- I worked on combining our visualizations into one dashboard.
- I simply copied our visualizations onto one combined sheet and adjusted the visualizations to fit the page while still being visible. I adjusted the colors to look presentable and clear to follow.

Gui:

- Created first draft file for mid-semester client presentation by combining the content from previous decks

Sprint 5 (April 21 - May 2):

Rachel:

- Completed the DATASETDOC.md for all questions and the dataset sets tree canopy metrics, tree canopy change metrics, and neighborhood tree canopy metrics
 - Submitted a Pull request to add the new file
- Edited code notebook in Github to be more organized by deleting redundant code and removing files previously generated to check outputs
- Started working on an ArcGIS Pro file to merge the table of classifications from my python notebook in the column "ownership" that labels parcels as 'city owned', 'non-city owned', and 'ROW' with the geospatial file defining neighborhood boundaries
 - Uploaded the tables of classified parcels to ArcGIS (only has PID and ownership columns)
 - Uploaded the geospatial Tree Canopy Metrics layer
 - Uploaded the geospatial Neighborhoods Tree Canopy Metrics layer
 - Used a "Add Join" tool to merge the table and geospatial metrics layer for the tree canopy metrics
 - Worked with Amelia to run the spatial join due to it requiring over 30 minutes to run

- Did the spatial join using the geospatial layer generated by the join and the neighborhoods layer. The spatial join assigns parcels to a neighborhood it has the greatest overlap with.
- I dropped the ROW polygon because it cannot be contained to any one neighborhood
- I used a spatial join to generate a new table where all parcels are labeled with the neighborhood they are in
- The next step is to aggregate the metrics to answer the question of how much tree canopy in each neighborhood is considered city-owned vs non-city owned

Amelia:

- Outlined Dataset Tree_Canopy_Height_Change_2014_to_2019 in detail in DATASETDOC.md for GitHub
 - After investigating CSV file using python (pandas), found that CSV contained 1,048,575 records
 - Documented number of instances as 4,560,574 as appearing in the shapefile in ArcGIS pro since that was more relevant to answering this question
- Assisted Rachel in performing a Spatial Join for her question
- Working on answering Question 6: “How do tree canopy height changes (2014-2019) compare between public and residential areas?”
 - Located dataset [Tree Canopy Height Change 2014 to 2019](#) on Analyze Boston
 - For now, prospectively, question 6 can be answered limitedly due to issues defining public vs. residential areas
 - Working on answering how tree canopy height has changed from 2014 to 2019 by ownership (ROW, non-city owned, and city owned)
- Question 6 process
 - Able to upload shapefile to ArcGIS pro
 - Looking to overlay with Canopy_Change_Assessment3A_Tree_Canopy_Metrics
 - Attempted to merge the two using the Spatial Join tool (>2 hrs process)
 - Facing blocker after Spatial Join did not give outcome I was hoping - was giving all null values for the Change Assessment
 - Attempting to troubleshoot and rerun but Spatial Join takes a very long time with the Height shapefile containing over 4.5 million records
 - Attempted Spatial Join (Target: Height; Join: Change; Join Operation: one to one; Match Option: Intersect) (2 hours 38 mins 9 secs), ran successfully
 - Created calculated field “HeightDiff” to take the height difference from 2014 to 2019 (!CanHghtT2! - !CanHghtT1!)
 - Ran Summary Statistics tool to get a sense of what the spread of the HeightDiff columns are per ownership type; found that there seem to be 244 records with ownership as <Null>

Grace:

- Learning curve with ArcGIS. Used google colab.
- Decided to use tree canopy change metrics dataset, percentage of trees in neighborhoods, ej communities dataset:

- Dissolved the ct into 17 neighborhoods. Statistically compared the ej categories that measured population definitions. Choose “LEP” “LowIncome” “Disability” in order to create a social score.
- Created a composite vulnerability index of $SV_key_nb = LEP_pct + Low\ Income$ because those two had the most negative scores.
- Found that places with more social score had less tree canopies.
- More can be seen in my comments in my colab notebook and arcgis.
- Still need to upload arcgis, colab notebook, and data documentation on github.
- Link to my arcgis:
<https://bucas.maps.arcgis.com/apps/mapviewer/index.html?webmap=8297c87ecaa7441394172e5aee97b4ef>

Gui:

- Created the first draft of the README.md file on Github and created a pull request (04/23), to be slightly updated for final submission the week later ([Notion Card](#))
 - README file included project overview, client questions we are answering, data sources, key findings and analyses and blockers faced
 - Julissa (TPM) approved the merge on Github on 04/24
- Started working on final client presentation
- Worked on my portion of the final report focusing on answering the question “How does tree canopy coverage vary across neighborhoods and by land use types” and future insights as part of the conclusion
- Along with the team, helped create the 2-slide poster that will be used on Demo Day, specifically in writing the methodology and goal of the project
- Started working on the 4-slide presentation that summarizes our project that will be presented in the final DS539 class

Talya:

- I had an issue extracting the data (as CSV or XLSX) from my ArcGIS file and faced a blocker there. After speaking with teammates, we found a way to fix it. Using the analysis tab and the Extract data tool, I was able to save the file as a CSV and then do my calculations.